

Transforming Human-AI Collaboration using “Large Whatever Models” (LWMs)

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Abstract

This Special Interest Group (SIG) focuses on re-imagining Human-AI collaboration in the age of off-the-shelf generative AI technologies reflecting the growing integration of AI into daily life and societal systems. The SIG addresses human-AI collaboration at three levels: interaction, collaboration, and symbiosis. Key topics include designing systems that adapt to human contexts, re-imagining collaboration interfaces, fostering inclusive urban environments, and balancing technological governance with equitable access. We also pose the question: what is the next big thing after GenAI to enhance collective intelligence, and societal equity and empower people? Our goal here is to build a sustainable, specialized, and interdisciplinary community beyond the scope of the SIG that focuses on creating a sustainable, equitable global impact through augmenting human capabilities with efficient, safe, and trustworthy collaborations with AI systems.

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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction paradigms*; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Generative AI, Large Language Models, Large Multimodal Models, ChatGPT, Human-AI Collaboration, Hybrid Intelligence, Symbiosis

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1 Introduction and Motivation

Human-AI collaboration is a research theme that is gaining strong traction in the Human-Computer Interaction (HCI) research community. The integration of artificial intelligence (AI) into everyday technologies through the spike in generative AI tools in the last

three years (e.g. large language models like ChatGPT¹, visual generation models like DALL-E 3², Stable Diffusion³, and Midjourney⁴, and multimodal models like Gemini⁵) has transformed how humans interact and think about AI and brought the conversation outside the research labs to the laymen in their everyday lives. The prevalence of generative AI technologies (or "Large Whatever Models", LWMs [13]) have enabled the wide creation of systems that can understand – or give the illusions of understanding – context, interpret human intentions, and provide adaptive support.

In this SIG, we discuss human-AI collaboration, which is a central subtopic in the research of interactive AI. Here we refer to three levels of the relationship: 1) interaction, 2) collaboration, and 3) symbiosis. *Interaction* refers to studying AI methods that "understand" people and can anticipate the consequences of their actions on people, and communicate their purposes to ground collaboration. This level also involves seeking more natural ways to communicate with AI, including multimodality, conversational UIs, and augmented reality (AR) interfaces. At the *collaboration* level, we consider concepts like cooperation, emotional intelligence, collective intelligence, and group work. At the *symbiosis* level, we study emergent properties of AI systems where people and AI combine their processes, skills, and experiences to achieve something greater together than just by themselves.

Human-AI collaboration provides gains in several application areas. For example, AI systems can assist clinicians by analyzing complex datasets, offering diagnostic insights, and even suggesting treatment plans in the healthcare domain (e.g. [18, 23, 26]). In education, AI-powered tools can adapt learning materials to individual student needs, fostering personalized education (e.g. [4, 6, 8]). Collaborative AI systems in creative industries, such as design or music, enable co-creation by combining human ingenuity with AI-generated suggestions (e.g. [7, 20, 21]). GenAI can also support and expedite research processes with limitations on what it can do for us [5, 13, 24]. Such systems not only extend human capabilities but also provide innovative solutions to complex problems.

In such a pivotal moment of scientific history, core philosophical questions like *is human-ai collaboration augmenting the skills of the human race or diminishing our capabilities?*, *How can we ensure equitable access to AI technologies?*, *How do we address concerns about job displacement and skill erosion?* become essential. Answering these questions requires interdisciplinary efforts from researchers, policymakers, and practitioners and here comes our role in CHI as a core interdisciplinary community of Human-Computer Interaction research. HCI researchers bring user-centric perspectives to AI design, ensuring that systems are not only functional but also intuitive and accessible. Furthermore, HCI research explores innovative interaction paradigms, such as multimodal communication and adaptive interfaces, that facilitate more natural and effective collaboration. We also have the technical skills to investigate the societal impact of those fast-growing and developing technologies and offer transitional solutions that help people use the technology effectively and empower them rather than replace them.

While there have been several SIGs in the last three years targeting aspects of AI/GenAI design, none specifically focused on the aspect of *human-AI collaboration*. In CHI'24, four SIGs were proposed targeting privacy protection [19], policies' impact on AI usage [9], AI for accessibility [10], and supporting research productivity [13]. In CHI'23, there was only one relevant SIG about responsible AI [25] and there were none in CHI'22. We think it is time to collect the interested but dispersed community and create a more substantial and specialized collaboration entity around Human-AI Collaboration, such as the HHAI: Hybrid-Human Artificial Intelligence conference [1], special journal issues, and sustainable interest groups [2] with larger research bodies such as (but not limited to) ADRA: The European AI, Data, and Robotics Association [3]. This SIG will build upon our extensive efforts over the past years in organizing events that reimagine Human-AI collaboration (e.g. [11, 12, 14–17, 22]), serving as a platform to discuss the next steps for our international community to sustain these efforts and create a global impact.

2 SIG Topics and Open Questions

The following topics are proposed as examples to guide the discussions within this SIG. Our goal is to identify which topics are a priority to discuss and resolve to enable *efficient, safe, and trustworthy collaborations between humans and AI systems*.

Topic 1: Collaboration Goal What defines the goal of human-AI collaboration: simple assistance, dynamic partnerships, or a true symbiosis? How can these collaborations evolve to not just augment human capabilities but create outcomes greater than the sum of their parts?

Topic 2: Common Grounds between Humans and AI How do humans and machines continuously adapt to each other and the context? How can machines understand their impact on humans before taking action? How do we design self-aware systems that can monitor and self-diagnose their interactions with the environment and other humans to self-improve their interactions? How can humans efficiently communicate their needs to AI systems?

Topic 3: AI Form and Collaboration Experience Do people react to physical embodiments of smart systems such as robots the same way they like non-tangible versions like software products? What are the pros and cons of embodied smart systems? How can these innovations support sustainable, inclusive, and context-sensitive human-AI collaborations? What are the emerging paradigms for effective and natural interactions between humans and AI systems? How do passive ambient intelligence systems differ in their societal impacts from active intelligence, and which applications suit each approach?

Topic 4: Ownership and Long-Term Impact on Humans What are the ethical responsibilities for the outputs generated by AI systems? How can we support people in taking responsibility for generated content when they do not know the source? How can we ensure these systems augment human skills rather than diminish them, and what safeguards are necessary to prevent societal disruptions, such as job displacement?

¹ChatGPT: <https://chat.openai.com>

²DALL-E 3: <https://openai.com/dall-e-3>

³Stable Diffusion: <https://stablediffusionweb.com>

⁴Midjourney: <https://www.midjourney.com>

⁵Gemini: <https://deepmind.google/technologies/gemini/#introduction>

Topic 5: Urban Inclusion How can AI systems be designed to promote inclusivity in urban environments? What role can multimodal understanding of diverse, real-world environments play in fostering accessibility, equity, and engagement for all citizens?

Topic 6: Collaboration Governance What are the global implications of governance structures like the AI Act on developing and adopting AI technologies supporting Human-AI Collaboration? How can political sovereignty and equitable access be ensured for technology owners worldwide?

Topic 7: The Future Beyond Large Language Models What is the next big technological breakthrough to support human-AI collaboration after Large Language Models (LLMs)? What global application areas will emerge as pressing needs for human-AI collaboration in the coming decades?

3 SIG Goal, Structure and Outcome

This SIG is a first step to starting a specialized community focusing on the unique challenges of Human-AI collaboration. As a first step, we aim to initiate a topics group [2] in ADRA [3] to have an existing body for creating and disseminating activities related to Human-AI collaboration. We will also submit a poster in a relevant scientific conference summarizing the results of the SIG. CHI has a good critical mass of authors who are relevant to the topic. For example, a quick search in CHI 2024 program using the term “collaboration” already yields 120+ contributions, 166+ contributions for the term “LLMs”, and there were two fully dedicated sessions in the late-breaking work track titled “Human-AI collaboration”. This gives a signal that such works might be further developing and maturing to the paper track later. The content of this SIG is relevant to both the academic and industrial audience doing research. Capitalizing on the positive feedback and the success of our SIG last year [13], we will follow a similar format.

The session is divided into two parts: The first part is a pitching format where prominent experts of the field pitch 5-7 minutes of core challenges related to Human-AI collaboration using GenAI and LWMs. The second part has interactive activities focused on getting the community to think together about the SIG topics. We will dedicate time to discussing what official bodies (e.g. conferences, journal special issues, workshops, etc.) could be used to expand and maintain the community. For content preservation beyond the SIG, we will create a website to publicize the SIG content and host all the post-SIG resources. We will also proactively approach workshop organizers, paper authors, and SIGs from the last three years and invite them to join the SIG.

The outcome is expected to be a vibrant community within CHI and across related bodies and organizations that can together have a lasting impact on future research and societally efficient activities in human-AI collaboration.

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References

- [1] 2022. *Hybrid-Human Artificial Intelligence Conference Series*. <https://hhai-conference.org/> Accessed: December 4, 2024.
- [2] 2024. *ADRA Topic Group*. <https://adr-association.eu/topic-groups> Accessed: December 4, 2024.
- [3] 2024. *The AI Data Robotics Association*. <https://adr-association.eu/> Accessed: December 4, 2024.
- [4] Mohammad Abolnejadian, Sharareh Alipour, and Kamyar Taeb. 2024. Leveraging ChatGPT for Adaptive Learning through Personalized Prompt-based Instruction: A CS1 Education Case Study. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–8.
- [5] William Agnew, A Stevie Bergman, Jennifer Chien, Mark Diaz, Selim El-Sayed, Jaylen Pittman, Shakir Mohamed, and Kevin R McKee. 2024. The illusion of artificial inclusion. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–12.
- [6] Sahan Bulathwela, María Pérez-Ortiz, Emine Yilmaz, and John Shawe-Taylor. 2022. Power to the learner: Towards human-intuitive and integrative recommendations with open educational resources. *Sustainability* 14, 18 (2022), 11682.
- [7] Luca Casini, Nicolas Jonason, and Bob Sturm. 2024. Sparks of Musical AGI? Challenges and perspectives in music co-creation with LLMs: A qualitative exploration of the music knowledge of LLMs and their use for music creation. In *International Conference on AI and Musical Creativity (AIMC) 2024*.
- [8] Alan Y Cheng, Meng Guo, Melissa Ran, Arpit Ranasaria, Arjun Sharma, Anthony Xie, Khuyen N Le, Bala Vinaithirathan, Shihe Luan, David Thomas Henry Wright, et al. 2024. Scientific and Fantastical: Creating Immersive, Culturally Relevant Learning Experiences with Augmented Reality and Large Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–23.
- [9] Marios Constantinides, Mohammad Tahaei, Daniele Quercia, Simone Stumpf, Michael Madaio, Sean Kennedy, Lauren Wilcox, Jessica Vitak, Henriette Cramer, Edyta Paulina Bogucka, et al. 2024. Implications of Regulations on the Use of AI and Generative AI for Human-Centered Responsible Artificial Intelligence. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–4.
- [10] Aayushi Dangol, Yun Huang, Srirangaraj Setlur, Adele Smolansky, Hariharan Subramonyam, Hyewon Suh, Jinjun Xiong, and Julie A Kientz. 2024. AI-Driven Support for People with Speech & Language Difficulties. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–4.
- [11] Passant Elagroudy. 2024. *CHI'24 SIG on Transforming HCI Research Cycles using Generative AI and "Large Whatever Models" (LWMs)*. <https://www.humane-ai.eu/event/lwms-hci/> Accessed: December 4, 2024.
- [12] Passant Elagroudy, Agnes Grünerbl, Giulia Barbareschi, Jan Spilski, Kai Kunze, Thomas Lachmann, and Paul Lukowicz. 2024. mobiCHAI-1st International Workshop on Mobile Cognition-Altering Technologies (CAT) using Human-Centered AI. In *Adjunct Proceedings of the 26th International Conference on Mobile Human-Computer Interaction*. 1–5.
- [13] Passant Elagroudy, Jie Li, Kaisa Väänänen, Paul Lukowicz, Hiroshi Ishii, Wendy E Mackay, Elizabeth F Churchill, Anicia Peters, Antti Oulasvirta, Rui Prada, et al. 2024. Transforming HCI Research Cycles using Generative AI and “Large Whatever Models” (LWMs). In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–5.
- [14] Passant Elagroudy and Paul Lukowicz. 2024. *Panel on Harnessing Generative AI for Inclusive Global Education in AIDA Symposium*. <https://www.vision4ai.eu/community-workshop-2024/#day2> Accessed: December 4, 2024.
- [15] Passant Elagroudy and Paul Lukowicz. 2024. *Workshop:EuroGen - Mapping the Future with Generative AI*. <https://www.vision4ai.eu/community-workshop-2024/#day1> Accessed: December 4, 2024.
- [16] Passant Elagroudy, Siobhan O'Neill, Anastasiya Zakreuskaya, Joanna Sorysz, Stefan Gerd Fritsch, Mohamed Selim, Marco Hirsch, Sebastian Vollmer, Georgios Spathoulas, and Paul Lukowicz. 2024. *LWM Hackathon: Enhancing Research Productivity*. <https://www.humane-ai.eu/event/lwmathon-research/> Accessed: December 4, 2024.
- [17] Hend Eldamaty, Passant Elagroudy, Jakob Karolus, Siobhan O'Neill, Joanna Sorysz, and Paul Lukowicz. 2024. *Future-Forward Families: Hacking AI-Powered Startups for Modern Parenthood*. <https://www.humane-ai.eu/event/future-forward-families/> Accessed: December 4, 2024.
- [18] Hamraz Javaheri, Omid Ghamarnejad, Paul Lukowicz, Gregor Alexander Stavrou, and Jakob Karolus. 2024. Design and Clinical Evaluation of ARAS: An Augmented Reality Assistance System for Open Pancreatic Surgery. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 376–385.
- [19] Tianshi Li, Sauvik Das, Hao-Ping Lee, Dakuo Wang, Bingsheng Yao, and Zhiping Zhang. 2024. Human-Centered Privacy Research in the Age of Large Language Models. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–4.

- [20] David Chuan-En Lin, Hyeonsu B Kang, Nikolas Martelaro, Aniket Kittur, Yan-Ying Chen, and Matthew K Hong. 2024. Inkspire: Sketching Product Designs with AI. In *Adjunct Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–6.
- [21] Sijia Liu, Ray Lc, Kexue Fu, Qian Wan, Pinyao Liu, and Jussi Holopainen. 2024. Dreamscaping: Supporting Creativity By Drawing Inspiration from Dreams. In *Proceedings of the 16th Conference on Creativity & Cognition*. 93–99.
- [22] Florian Müller, Albrecht Schmidt, Clara Sayffaerth, Passant Elagroudy, and Yuanting Liu. 2024. *The AI and I Hackathon*. <https://www.fortiss.org/en/events/ai-and-i-hackathon> Accessed: December 4, 2024.
- [23] Dimitrios P Panagoulas, Filippas A Palamidas, Maria Virvou, and George A Tsihrintzis. 2023. Evaluating the potential of LLMs and ChatGPT on medical diagnosis and treatment. In *2023 14th International Conference on Information, Intelligence, Systems & Applications (IISA)*. IEEE, 1–9.
- [24] Albrecht Schmidt, Passant Elagroudy, Fiona Draxler, Frauke Kreuter, and Robin Welsch. 2024. Simulating the Human in HCD with ChatGPT - Redesigning Interaction Design with AI. *Interactions* (Jan 2024).
- [25] Mohammad Tahaei, Marios Constantinides, Daniele Quercia, Sean Kennedy, Michael Muller, Simone Stumpf, Q Vera Liao, Ricardo Baeza-Yates, Lora Aroyo, Jess Holbrook, et al. 2023. Human-Centered Responsible Artificial Intelligence: Current & Future Trends. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–4.
- [26] Ehsan Ullah, Anil Parwani, Mirza Mansoor Baig, and Rajendra Singh. 2024. Challenges and barriers of using large language models (LLM) such as ChatGPT for diagnostic medicine with a focus on digital pathology—a recent scoping review. *Diagnostic pathology* 19, 1 (2024), 43.